

THE STORY OF ROBOTICS

ROBOTS HAVE DEVELOPED FROM BASIC MECHANICAL TOOLS AND TOYS TO PLAYING AN IMPORTANT ROLE IN MODERN SOCIETY

Robots are built in a wide range of forms, and there is no single definition of a robotic device that encompasses all its aspects. However, the vast majority are electromechanical machines that can perform tasks and manipulate their environment in accordance with a set of preestablished instructions.

Robots vary hugely in form and function. Physically, they range from a jointed arm with tools at the end to the human-shaped “android” beloved of science fiction enthusiasts. Operationally, they are just as diverse and include machines whose function is defined by their form—such as ancient Egyptian water clocks, or clepsydra, and Jacquard looms that could weave textile patterns based on instructions stored on punched cards. There are also robots that are versatile devices capable of performing a wide range of tasks and reacting to external stimuli, such as robot space probes.

MODERN ROBOTS

The word “robot” was first used by Czech writer Karel Čapek in his 1920 science fiction play

R.U.R. (Rossum’s Universal Robots), and derives from the Czech term for forced labor—indeed, most robots do perform their tasks by obeying instructions either embedded in their design, or passed on to them by a governing computerized controller in the form of software. Such robots excel at carrying out repetitive, complex, or detailed tasks at high speed and without fatigue. More versatile robots, such as the da Vinci surgical robot, often function under the direct command of a human operator—a technique known as telepresence, which may use cameras or more complex technology to relay the robot’s “view” of a hostile, dangerous, or complex environment. Increasing numbers of robots are being equipped with artificial intelligence, which allows them to make decisions of their own.



ARTIFICIAL INTELLIGENCE

For many robotics enthusiasts, the ultimate goal of computing is to develop a form of intelligence capable of mimicking human behavior. Most forms of artificial intelligence (AI) used in robots so far have involved the machine recognizing and reacting to scenarios according to a set of preprogrammed rules that its computer can understand. AI research has made its greatest advances in applications such as chess computers.

“ROBOTICS HAS BECOME A SUFFICIENTLY WELL DEVELOPED TECHNOLOGY TO WARRANT ARTICLES AND BOOKS ON ITS HISTORY AND I HAVE WATCHED THIS IN AMAZEMENT, AND IN SOME DISBELIEF, BECAUSE I INVENTED IT. NO, NOT THE TECHNOLOGY; THE WORD.”

Isaac Asimov, US writer, from *Counting the Eons*, 1983

c.250 BCE

Early robotics

Automated machinery, such as Egyptian water clocks, are considered by some to be an early form of robotics.



Ctesibius's clepsydra

1801

Programmable loom

The Jacquard loom, a programmable device that can weave textile patterns automatically, is developed.



Jacquard loom

1942

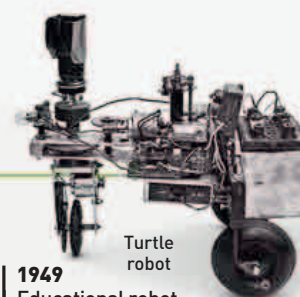
Asimov's laws

In his book *I, Robot*, US writer Isaac Asimov devises three laws to govern his fictional robots. These go on to influence real-life robot builders.

1949

Educational robot

The highly popular Turtle design developed by William Grey Walter is equipped with a variety of sensors that allow it to react to its environment.



Turtle robot

c.1206

Al-Jazari's automata

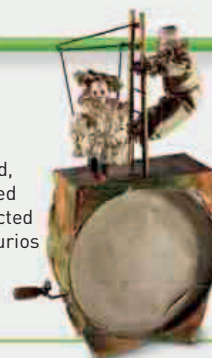
One of Arabic inventor al-Jazari's many achievements is a band of automata that can be commanded to play different pieces of music.



Al-Jazari's band

19th century Mechanical toy

From the European Renaissance onward, robotic figures, called automata, are collected as playthings and curios by the wealthy.



Drum automaton

1961

Industrial robot

The first commercial manufacturing robot, a robotic arm called Unimate, enters service at a General Motors plant in the US.



Robotic arm